

Wing dihedral affects the roll stability derivative of an airplane because it causes the lift on the right and left semispans to respond differently to sideslip. As shown in Fig. 5.6.2, on a wing with positive dihedral, sideslip produces an increase in angle of attack on the windward semispan and a decrease in angle of attack on the lee-side semispan. The sideslip component of relative wind is denoted in Fig. 5.5.2 as v . When this is resolved into components parallel with and normal to each semispan, as shown in Fig. 5.6.2, the upward normal component is $v \sin \Gamma$ on the right semispan and $-v \sin \Gamma$ on the left semispan. For small aerodynamic angles, the sideslip velocity can be closely approximated as the product of the airspeed and sideslip angle, $v \cong V\beta$. Likewise, the semispan angle of attack is closely approximated as the upward normal component of relative wind divided by the airspeed. Thus, sideslip produces a change in angle of attack that can be closely approximated for the right and left semispans, respectively, as

$$(\Delta\alpha)_{\text{right}} = \frac{v \sin \Gamma}{V} = \frac{V\beta \sin \Gamma}{V} = \beta \sin \Gamma \quad (5.6.2)$$

$$(\Delta\alpha)_{\text{left}} = \frac{-v \sin \Gamma}{V} = \frac{-V\beta \sin \Gamma}{V} = -\beta \sin \Gamma \quad (5.6.3)$$

The differential in angle of attack between the right and left semispans creates a differential in lift that produces a rolling moment. From Fig. 5.1.1, we see that positive lift on the left semispan ($y_b < 0$) produces a positive rolling moment, while positive lift on the right semispan ($y_b > 0$) produces a negative rolling moment. Thus, the rolling moment arm about the junction of the two semispans for the lift on any wing section is $-y_b / \cos \Gamma$ and the differential length measured perpendicular to the wing sections is $dy_b / \cos \Gamma$. Using the approximation for small sideslip angles given by Eqs. (5.6.2) and (5.6.3), the net rolling moment that results from sideslip combined with wing dihedral can be written

$$\begin{aligned} (\Delta C_\ell)_{\Gamma w} &= \int_{y_b = -b_w/2}^0 \frac{\tilde{C}_{L_i, \alpha} c_w}{S_w b_w} (-\beta \sin \Gamma) \frac{-y_b}{\cos \Gamma} \frac{dy_b}{\cos \Gamma} + \int_{y_b = 0}^{b_w/2} \frac{\tilde{C}_{L_i, \alpha} c_w}{S_w b_w} (\beta \sin \Gamma) \frac{-y_b}{\cos \Gamma} \frac{dy_b}{\cos \Gamma} \\ &= -\frac{\beta \sin \Gamma}{S_w b_w \cos^2 \Gamma} \int_{y_b = -b_w/2}^{b_w/2} \tilde{C}_{L_i, \alpha} c_w |y_b| dy_b \end{aligned} \quad (5.6.4)$$

where S_w and b_w are the reference area and reference length, respectively, and $\tilde{C}_{L_i, \alpha}$ is the *local in situ section lift slope*, including the effects of local induced downwash.

The spanwise variation of local induced downwash is affected by the sideslip. Thus, the derivative of Eq. (5.6.4) with respect to β must be written as

$$\frac{\partial}{\partial \beta} (\Delta C_\ell)_{\Gamma w} = -\frac{\sin \Gamma}{S_w b_w \cos^2 \Gamma} \left(\int_{y_b = -b_w/2}^{b_w/2} \tilde{C}_{L_i, \alpha} c_w |y_b| dy_b + \beta \int_{y_b = -b_w/2}^{b_w/2} \frac{\partial \tilde{C}_{L_i, \alpha}}{\partial \beta} c_w |y_b| dy_b \right)$$

However, the roll stability derivative, $C_{\ell, \beta}$, is evaluated at $\beta = 0$, and the contribution of wing dihedral to this aerodynamic derivative can be written